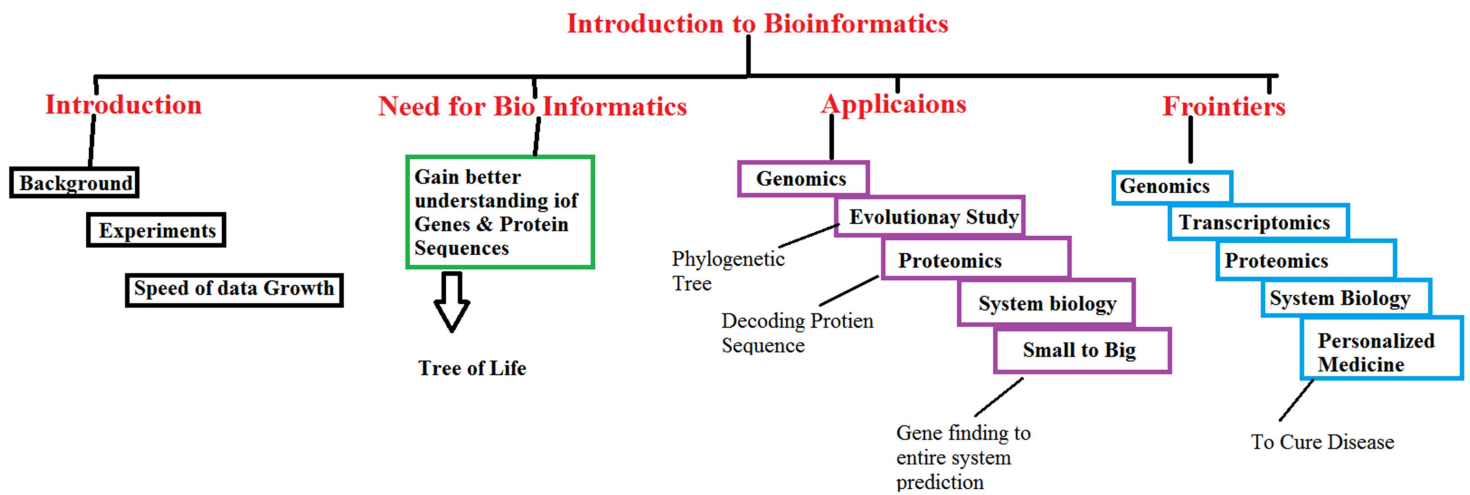


BIF401 Bioinformatics-1

Grand Quiz Preparation(Part 1)

Chapter 1- Introduction to Bioinformatics



• BACKGROUND

1. Bioinformatics is an **interdisciplinary science** at the cross-roads of **biology, mathematics, computer science, chemistry and physics**.
2. Human genome has over **25,000 genes**
3. The genes may produce **> 250,000** different proteins

• EXPERIMENTS IN BIOLOGY

These instruments include:

1. **Next Generation Sequencers (NGS)** for whole genome sequencing
2. **High Resolution Mass Spectrometry** for whole proteome profiling
3. **Nuclear Magnetic Resonance Spectroscopy** for structural studies

• DIGITALIZATION OF BIOLOGY

Produced data is stored on **Computer disks**. The data may include **text, numbers, symbols or images**.

• SPEED OF DATA GROWTH

1. Data is being accumulated at **exponentially increasing rates**
2. **Doubling** every few years

MOTIVATION

It is an **interdisciplinary field** as it covers the information of biological digital information including human, plants, animals and microorganisms.

It demands a **very low cost infrastructure** and **hardly any lab equipment**.

SCOPE OF BIOINFORMATICS

Bioinformatics primarily deals with digitalized biological information

ACTIVITIES IN BIOTECHNOLOGY

Developing algorithms, writing software, Statistical evaluation of data etc

NEED FOR BIOINFORMATICS

The need for bioinformatics is on a rapid rise as biological **data is rapidly increasing** and becoming available online, free of any cost.

Informatics and **Bio-science** are the **umbrella terms** given to a set of allied disciplines which make up the field.

Quote

Biology easily has 500 years of exciting problems to work on.

Donald Knuth, Professor, Stanford University

“tree of life”

which consist of:

➤ Bacteria

➤ Archaea

➤ Eucarya

- From **gene sequences** to **protein sequences** Bioinformatics is the way forward
- **Protein structure**, protein-protein interactions and systems biology are other research areas in bioinformatics

APPLICATIONS OF BIOINFORMATICS

GENOMICS

Bioinformatics can help in assembling

- DNA sequencing data.
- Gene Finding
- Genome Assembly
- Variation in Genomes
- Transcription Data
- Databases

EVOLUTIONARY STUDIES

- Evolutionary relationships
- Evolutionary distances
- **Phylogenetics**
- **Tree of life**

PROTEOMICS

Bioinformatics can help us in **decoding protein sequences**.

- **Systems Biology**
- **Personalized Medicine**

APPLICATIONS OF BIOINFORMATICS

- **Genomics**
- **Transcriptomics**
- **Proteomics**
- **Metabolomics**
- **Structural Proteomics**

- Drug Design
- Systems Biology
- Personalized Medicine

FRONTIERS (boundaries) IN BIOINFORMATICS

Frontier in Bioinformatics includes

- Next generation genomics
- Transcriptomics
- Proteomics

FRONTIER IN GENOMICS

To sequence the **whole genome** with the bioinformatics tool of Next generation sequencing (NGS)

- Next generation sequencing of whole genomes
- Massive amounts of data (Tera byte files)

FRONTIER IN TRANSCRIPTOMICS

- to **identify** those **matters** which are **unknown** yet or under discussion
- Identification of **hitherto**(until now) **unknown types** as well as roles of **RNA**

FRONTIER IN PROTEOMICS

- **Identification** of **low abundance proteins** in patient tissue samples
- Large scale protein identification
- Identification of **known** as well as novel post-translational modifications

FRONTIER IN PROTEIN STURCTURE

To understand the **layer folding of proteins** that how they are processed,

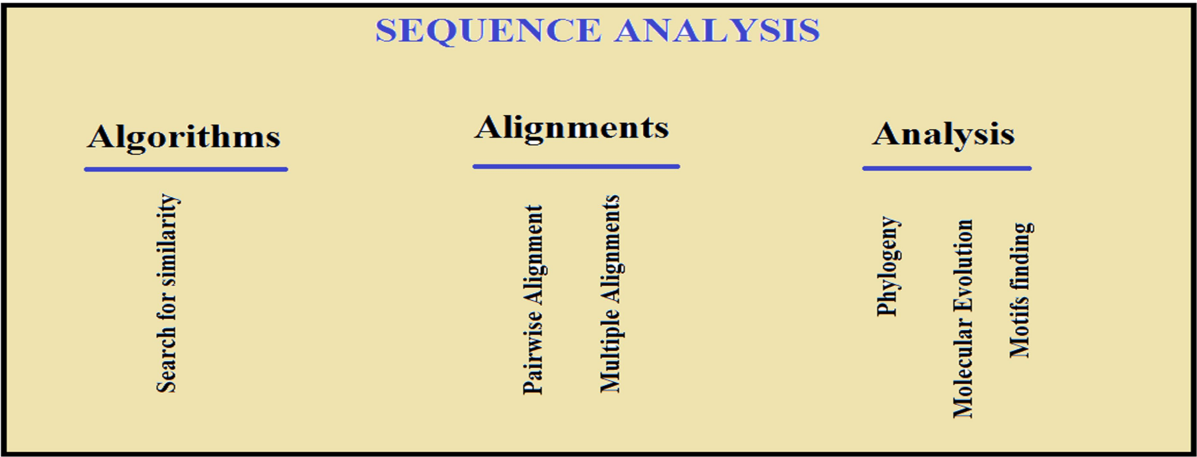
FRONTIER IN SYSTEM BIOLOGY

To understand **Cells function** as a whole!

FRONTIER IN PERSONALIZED MEDICINE

- Personalize the medicine for **exact cure** of a disease.
- Some medicines have **side effects** on certain person

Chapter 2 - Sequence Analysis



Gene, mRNA and Protein Sequences

Central Dogma

Transfer of this information from DNA to these molecules is termed as “**Central Dogma**”

- DNA codes for an RNA
- RNA in turn helps produce Proteins
- Proteins along with some other molecules form cells including their organelles and membrane

DNA-> RNA ->Protein.

DNA sends the **information to cell** via **mRNA** and that **sequence the amino acids** according to coded information and **protein structure** is formed and that protein forms a **cell**.

•DNA

DNA molecule is **double helix** structure contains **base pairs** composed of **nucleotides** and these nucleotides are composed of **sugar phosphate** group and are bind with each other with **hydrogen bonds**.

- DNA is constituted by **four bases**
- These are Adenine (A), Cytosine (C), Thymine (T), and Guanine (G)
- DNA molecule may contain thousands of A, C, T and G.
- **A pairs with T** only and **C pairs with G** with the help of **Hydrogen bonding**

•RNA

- RNA is constituted by **four bases**
- These are Adenine (A), Cytosine (C), **Uracil (U)**, and Guanine (G)
- RNA molecule may contain thousands of A, C, U and G.
- **A pairs with U** only and **C pairs with G**

Normally all the nucleotides are same in both DNA and RNA except one position in **RNA** which is **U (Uracil)** and in **DNA** it is **T (Thiamin)**

DNA molecule although is **double stranded** and RNA is **single stranded** but there is **difference** in **sugar composition**.

RNA has **Ribose sugar** and **DNA** has **de-oxyribose sugar**:

Adenine and Guanine collectively called **Purines** while Cytosine, Uracil, and Thymine are called as **Pyrimidine**.

TRANSCRIPTION

Transcription is the process by which the information in a **strand of DNA** is copied into a new molecule of **messenger RNA (mRNA)**.

NUCLEOTIDES

DNA and RNA molecule than these **are composed of four** other molecules which are named as **Nucleotides**

TRANSLATION

Cells are built of proteins and carbohydrates and these proteins are made in results of transformation of RNA molecule and this transformation is called as **translation**.

- RNAs code for proteins
- **Codons** of **three nucleotides** from the RNA molecules code one amino acid at a time
- This process takes place at the **ribosomes** and is called translation

AMINO ACIDS

In all, **20 different amino acids** can be coded by such codons during translation. The amino acids polymerize to form proteins

If we observe the **structure of amino acid** it contains nitrogen, hydrogen, oxygen and two carbon atoms and a **variable group R**.

- When polymerizations takes place **water is formed**
- **Polymerization** of Amino Acids by **Peptide Bond**

Amino acids are **joined** with each other with **peptide bonds** and fold with each other in **3D form** they make protein structure.

STORAGE OF BIOLOGICAL SEQUENCE INFORMATION

- DNA sequences may contain hundreds of thousands of bases
- RNA sequences can be a little less than the DNA but they are in a much larger variety

SOLUTIONS DATABASES

- Data bases for **DNA & RNA** the public database is **GenBank (by NIH)**.
- For **proteins** the public database is **UniProt (by Uniprot Consortium)**
- Both GenBank and UniProt are **online database** and the DNA, RNA and Protein sequences are available here **online for public** and researchers.

USING GENBANK

GenBank can be searched by:

- Sequence
- ID
- Name
- Species etc.

Other information includes:

- Locus
- Accession number
- Authors
- Journal etc

USING UNIPROT

UniProt is public database which is being used to search **the sequence of proteins**

In **home page** there is a box named "**Swiss Prot**" which contains **human curated protein** information

UniProt can be searched by:

- **Amino Acid Sequence**
- ID
- Name
- Species etc.

COMPARING SEQUENCES

By comparing sequences of DNA, RNA and Proteins we can get

- **Similarity**
- Specific **difference** due to some disease or mutation
- Some evolutionary **relationship**.

SIMILARITIES & DIFFERENCES IN SEQUENCES

If some sequences are exactly similar to each other it means there is some;

- Regular expression in **cell or system**
- Specific presence of **protein or gene**
- Similar nucleotide

Exact Matching

- Exact Matches include **complete matching in terms of residues** and their **same order**
- Exact matching includes **partial matching** in terms of residues and order with room for variations in both.

Pairwise Sequence Alignment

Alignment

The process of **inexact matching** while keeping in view the **conserved residues** is called sequence "Alignment"

Pairwise sequence alignment is therefore alignment of **a pair of two sequences**

Pairwise sequence alignment compares **two amino acid** or **nucleotide sequences**

Matches are colored and missing nucleotides are denoted by

There are **two types of pair alignments**.

1. **Global**

2. **Local**

- **Global alignment** - **maximizes** the number of matches between the query and source sequences along the **entire length** of both the sequences.
- **Local alignment** - gives the **highest scoring local match** between both query and sequences.

Optimal alignment - one that exhibits the most correspondences between the query and the source sequences. It is the alignment with the **highest score**.

Pairwise Sequence Alignment is used **to identify regions of similarity** that may indicate functional, structural and/or evolutionary relationships between two biological sequences (protein or nucleic acid)

PAIR WISE SEQUENCE ALIGNMENT

Insertions or deletions ("**indels**") result in **gaps** in alignments

- **Substitutions** result in **mismatches**

DOT PLOTS

- Dot plots employ **dot matrix** representation for **pairwise alignment** and comparison
- Sequences are written on **top & left side** of a dot matrix grid
- **Dot plots employ dot matrix with two sequences plotted on top and left of the matrix**

- **Matches** are represented by **dots**
- **Dots on diagonals** are connected and represent **alignments**

Dot plots provide us the **Global similarity** between the two sequences

Dot plot helps us to find the **threshold difference** among two sequences

EXAMPLE OF DOT PLOTS

Dot plot of the **human Cytochrome** and **Tuna Fish Cytochrome** than the diagonal alignment of sequence

IDENTITY VS SIMILARITY

There are two concepts for sequence analysis

1. Identity
2. Similarity

Sequence Identity

Identity means the **counting number** of nucleotides or amino acids which **exactly match** when two biological sequences are matched.

Formula for Identity:

Identity = No. of Matches / smaller length × 100

$$= \frac{5}{5} \times 100 = 100\%$$

- **Gaps are not counted**
- Identity measurement is **made on the shorter** of the two sequences

Similarity means the **comparison** between two different sequences calculated by **alignment approach**.

Sequences vary due to insertions, deletions and substitutions

APPROACHES TO ALIGN THE SEQUENCE

1. Global Alignment

In Global alignment we compare both sequence from **end to end completely**.

Global alignment creates **an end-to-end matching** between two sequences

2. Local Alignment

In **Local alignment** we compare **one whole sequence** with the **one portion of other**.

Local alignment focuses on the **highly matching sub regions** within the sequences

WHY LOCAL ALIGNMENT

- Local alignments have the **power to detect small regions of high similarity** between two sequences
- Such matches may be "**domains**" or "**motifs**" in case of proteins
- **DOMAIN SHUFFLING**

Aligned portions of sequence can be considered in **varying orders** and this process is called as **domain shuffling**.

ALIGNING, INSERTION & DELETION

Insertion means addition of **amino acids in protein** sequence and addition of **nucleotides in DNA** sequences.

And deletion means **removal of amino acids** from protein sequence and **removal of nucleotides** from DNA or RNA sequences.

Insertion of gap is called as **-ve or plenty**

Removal and addition of amino acids in proteins and nucleotides in DNA, RNA by using Gaps named as Indels

ALIGNING MUTATION IN SEQUENCES

Mutation is totally different from Indels, because in Mutation we **replace** the amino acid with other amino acids and replace the nucleotides with other and we don't use Gaps

CONCLUSION

In identity alignment we use Gaps and in mutation we use substitution penalties and penalties depend upon the substitution.

To find **matching in nucleotides and amino acids** of two sequences we **use dot plot** method.

But dot plot **cannot capture** the **insertions, deletions and gaps** in the sequences.

Modification in Dot Plot

We represent the matching nucleotides with **+1 while gaps**, substitutions, insertions and mutations can be represented as **-1 in dot plot**

INTRODUCTION TO DYNAMIC PROGRAMMING

Dynamic programming is an algorithmic technique used commonly in sequence analysis.

Dynamic programming is used when recursion could be used but would be inefficient because it would repeatedly solve the same sub problems.

Need of algorithm

Comparison of two sequences one by one it need time and is computationally expensive method. That's why we need algorithm

If we compare two sequences of length "**n**" than it would be "**n²**", its order is **O (n²)**

DYNAMIC PROGRAMMING METHODOLOGY

Dynamic programming (DP) helps **reduce the large computational cost** associated with sequence comparisons

DP uses a "**scoring function**" to deal with matches, mismatches and gaps

Conclusion

All the alignments are done in **diagonal way** in dot plot matrix. For **total score** we make calculations in diagonal way and after calculation **best one is selected**

Needleman Wunsch Algorithm

We Start with a **zero** in the second row, second column.

		Index: j = 0 : m				
		A	C	G	C	
Index: i = 0 : n		0	$-\gamma$	-2γ	-3γ	-4γ
	A	$-\gamma$	$C(i,j) = \max \begin{cases} C(i-1,j-1) + w(S_i, T_j) \\ C(i-1,j) - \gamma \\ C(i,j-1) - \gamma \end{cases}$			
	C	-2γ				
	A	-3γ				
	C	-4γ				

Move through the cells row by row, calculating the score for each cell.
 Left to right and top to bottom the best element (having high score) is selected.
The matrix is computed progressively until the bottom right element

- α Alpha = Match reward
- β Beta = Mismatch penalty
- γ Gamma = Gap penalty

BACKTRACKING ALIGNMENTS

To find an **optimal alignment** in Needleman Wunsch Algorithm we use traceback method

	λ	C	T	C	G	C	A	G	C
λ	0	-5	-10	-15	-20	-25	-30	-35	-40
C	-5	10	5	0	-5	-10	-15	-20	-25
A	-10	5	8	3	-2	-7	0	-5	-10
T	-15	0	15	10	5	0	-5	-2	-7
T	-20	-5	10	13	8	3	-2	-7	-4
C	-25	-10	5	20	15	18	13	8	3
A	-30	-15	0	15	18	13	28	23	18
C	-35	-20	-5	10	13	28	23	26	33

+10 for match, -2 for mismatch, -5 for space

After completely matrix calculations we apply traceback to find the optimal alignment and
 traceback starts from bottom right (maximum score) to top side

REVISITING LOCAL AND GLOBAL ALIGNMENTS

Traceback strategy allows us to **differentiate between a local and a global alignment**
“Traceback” is the technique by which we can **check the sequences from any end** of the matrix box.

- Dot plot** help us in finding matching **residues** of two sequences while **Needleman Wunsch** helps us to find the **global alignments**.
- If some sequences have **different regions** of nucleotides which **does not match to any other** for that alignment **we prefer Global alignment** not local, but that does not penalize leading or trailing end

MOVING FROM GLOBAL TO LOCAL ALIGNMENT

DNA has **coding** and **noncoding** regions.
Coding regions are called **“EXON”** expressed as protein and they remain more conserved due to their role in making functional proteins.
 And **noncoding regions** of DNA are called as **“INTRONS”** which are more likely involved in mutations than coding ones.
 It means **high degree of alignment** can be find among two **exons**.

Local Alignments can identify **exons** which are present in both sequences
Exons in DNA tend to be **more conserved** as compared with introns

SMITH WATERMAN ALGORITHM

In **global alignment** we compare the sequence from **end to end** but in **local alignment** we compare the **sequences in segments**

The Smith Freshman algorithm is different from Needle man.

- Top row and Colum are set to zero.
- Alignment can end anywhere.
- Traceback starts from highest score.

Local Alignments can identify coding portions in a DNA

EXAMPLE OF SMITH WATERMAN ALGORITHM

The only difference between Needleman and Smith Waterman is that zero “0” is placed in the relationship.

Example of Smith Waterman Algorithm

	λ	C	T	C	G	C	A	G	C
λ	0	0	0	0	0	0	0	0	0
C	0								
A	0								
T	0								
T	0								
C	0								
A	0								
C	0								

$C[i,j] = \max \begin{cases} C[i-1,j-1] + score(i,j) \\ C[i-1,j] - \gamma \\ C[i,j-1] - \gamma \\ 0 \end{cases}$

+1 for a match, -1 for a mismatch, -5 for a space

- **Traceback can start from any position in the scoring matrix**
- **Local alignments can be extracted by starting from a high score till reaching ‘0’**

REPEATED ALIGNMENTS

By making some change in strategy of traceback we can find the repeated sequences.

Two modifications

- **Use a score threshold ‘T’ above which the matches will be considered.**
- **This will help avoid low scoring local alignments**
- **Traceback should find multiple aligned regions by multiple traceback steps**

REPEATED ALIGNMENTS STEPS

- **Continue to traceback until you reach the top row again**
- **Repeat the previous two steps**
- **Stop the trace back if you reach ‘0’ at (0,0)**

Traceback Strategies:

We have seen how biological sequences can be searched and compared using various **recurrence relations** and **traceback strategies**

INTRODUCTION TO SCORING ALIGNMENTS

There are **two types of scoring alignments**;

- Optimal Alignments
- Best Alignment

MEASURING ALIGNMENTS SCORES

The matrix has **positive and negative scores** both, matches and mismatches therefore are all considered because it's a diagonal pattern

SCORING MATRICES

How to build scoring matrices?

- We analyze the observed frequency with which each amino acid / nucleotide is substituted by an other one in similar proteins/genes

Scores in the Scoring Matrices

- Scoring matrices may contain +ve and -ve values as well as 0.
- +ve values are for more frequent matches / mismatches
- -ve values are for unlikely one

There are two types of scoring matrices.

➤ PAM

➤ BLOSUM

- Protein sequences have a choice of PAM and BLOSUM matrices.
- Nucleotide sequences have choices for a pair of match/mismatch costs.

PAM MATRICES

PAM means “Point Accepted Mutation”

Point accepted mutations means the substitution of one amino acid in a sequence with another that protein function remains conserved.

PAM UNIT

PAM unit is actually that **time** during which **1% amino acid** undergo for **acceptable mutation**.

Important: If two sequences diverge by **100 PAM units**, it does **not** imply that they are **totally different** in each position

STEP TO COMPUTE PAM MATRICES

• STEP TO COMPUTE PAM MATRICES

1. Align the protein sequences which are 1-PAM Unit diverge.
2. Let A_{ij} be the number of times A_i is substituted by A_j .
3. Compute the frequency f_i of amino acid A_i .

Then, $PAM1 = \frac{A_{ij}}{\sum_k A_{ik}}$

$PAM 'n' = (PAM1)^n$

- PAM1, PAM2 ...PAM250 can be computed
- **PAM120** is considered **optimal** scoring matrix!

BLOSUM MATRICES

BLOSUM matrices can be used to align the protein sequences. BLOSUM matrices was first purposed in **1992** by Henikoff et al.

BLOSUM matrices is also called the **Block substitution matrix without any gap** although it has mismatches in sequences.

There are three steps to compute the BLOSUM Matrices.

Step 1: Eliminate sequences that are identical in $x\%$ positions

Step 2: Compute observed frequency $f_{i,j}$ of aligned pair A_i to A_j . Hence, $f_{i,j}$ becomes the probability of aligning A_i and A_j in the selected blocks.

Step 3: Compute f_i which is the frequency of observing A_i in the entire block

Typically used matrices: **BLOSUM62** or **PAM120** in PAMx, larger x detects more divergent sequences.

BLOSUM62 and **PAM120** are considered optimal for use in **protein sequence alignment**

For BLOSUMn, a lower n detects more divergent sequences

MULTIPLE SEQUENCE ALIGNMENT (MSA)

Multiple sequence alignment involves comparison of **three or more sequence**

In Multiple sequence Alignments we compare multiple number of protein and DNA sequences to identify the matches and mismatches.

MSA vs Pairwise

For pairwise alignments, we used Dynamic Programming

For MSA, Dynamic Programming may get very expensive, so we use Progressive Alignment

- Pairwise alignment was either local or global
- Multiple alignments are mostly global

APPLICATION OF MSA

- Predict secondary and tertiary structures of new protein sequences
- Evaluate evolutionary order of species or “Phylogeny”

METHODOLOGY

- Pairwise alignment is the alignment of two sequences
- MSA can be performed by **repeated application of pairwise alignment**

MSA can help align multiple sequences. Progressive alignment can help perform MSA. Need to remove sequences with **>80% similarity**.

PROGRESSIVE ALIGNMENT FOR MSA

- **Progressive alignments** are used in aligning multiple sequences
- **Iterative approaches** can help refine results from progressive alignments

SHORTCOMING OF THIS APPROACH

- Dependence upon initial alignments
- If sequences are dissimilar, **errors in alignment are propagated**
- Solution: **Begin by using an initial alignment**, and refine it repeatedly

Progressive Alignment for MSA

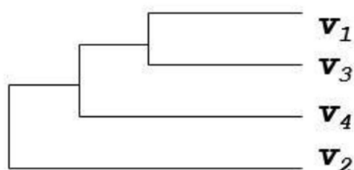
- Step 1 : **Pairwise Alignment** of all sequences
Example: S_1, S_2, S_3, S_4 , so that is 6 pairwise comparisons!
- Step 2: Construct a **Guide Tree (Dendogram)** using a *Distance Matrix*



- Step 3: Progressive alignment **following branching order in tree**

$Similarity = \frac{Matches}{Comparisons}$

	v_1	v_2	v_3	v_4
v_1	-			
v_2	.17	-		
v_3	.87	.28	-	
v_4	.59	.33	.62	-

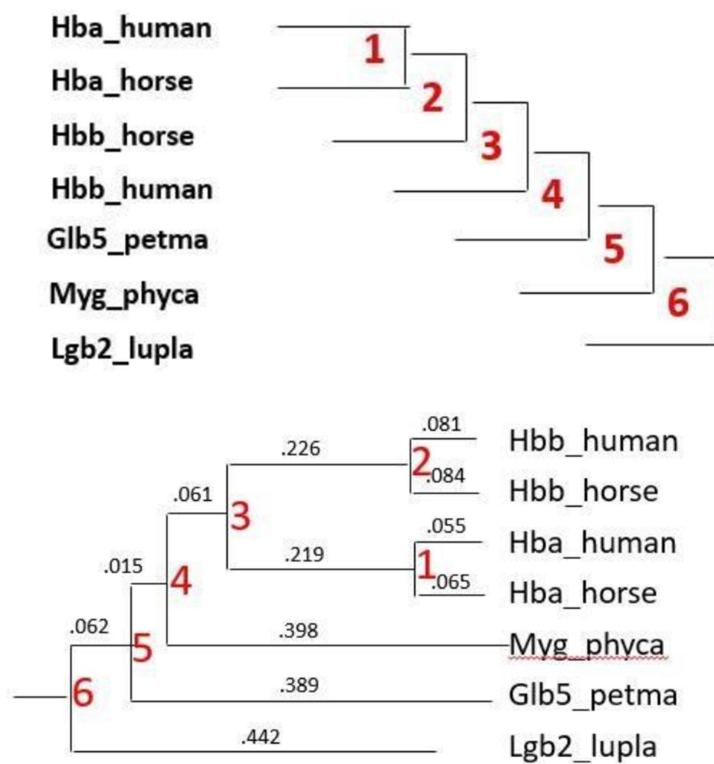


MSA-EXAMPLE

distance between 2 sequences = $1 - \frac{\text{No. identical residues}}{\text{No. aligned residues}}$

		Hbb_human	Hbb_horse	Hba_human	Hba_horse	Myg_phyca	Glb5_petma	Lgb2_lupla
Hbb_human	1	-						
Hbb_horse	2	.17	-					
Hba_human	3	.59	.60	-				
Hba_horse	4	.59	.59	.13	-			
Myg_phyca	5	.77	.77	.75	.75	-		
Glb5_petma	6	.81	.82	.73	.74	.80	-	
Lgb2_lupla	7	.87	.86	.86	.88	.93	.90	-

Progressive alignment using sequential branching



MSA can be better performed using **clustering strategies** followed by alignment of the alignments later. **CLUSTAL** is a **free online tool** that does all of this for us!

CLUSTALW

Developed by **European Molecular Biology Laboratory & European Bioinformatics Institute**
Performs alignment in:

- **slow/accurate**
- **fast/approximate**

SCOPE

- create multiple alignments,
- optimize existing alignments,
- profile analysis &
- create phylogenetic trees

CLUSTALW	MAFFT	PRRN
General Setting Parameters: Output Format: CLUSTAL Pairwise Alignment: ☒ FAST/APPROXIMATE ☐ SLOW/ACCURATE Enter your sequences (with labels) below (copy & paste): ☒ PROTEIN ☐ DNA Support Formats: FASTA (Pearson), NBRF/PIR, EMBL/Swiss Prot, GDE, CLUSTAL, and GCG/MSF Or give the file name containing your query Choose File No file chosen		

Using CLUSTALW

- CLUSTALW can use multiple file formats including: **EMBL/SwissProt, Pearson (Fasta)** etc

CLUSTAL Omega is now also available which includes several upgrades!

INTRODUCTION TO BLAST-I

National Center for the Biotechnology Information (NCBI) – USA
BLAST developed in 1990

“Basic Local Alignment Search Tool”

Searches databases for query protein and nucleotide sequences
Also searches translational products etc.

Online availability www.blast.ncbi.nlm.nih.gov/Blast.cgi
Basic BLAST

Choose a BLAST program to run.

<u>nucleotide blast</u>	Search a nucleotide database using a nucleotide query <i>Algorithms:</i> blastn, megablast, discontinuous megablast
<u>nucleotide blast</u>	Search a nucleotide database using a nucleotide query <i>Algorithms:</i> blastn, megablast, discontinuous megablast
<u>protein blast</u>	Search protein database using a protein query <i>Algorithms:</i> blastp, psi-blast, phi-blast, delta-blast
<u>blastx</u>	Search protein database using a translated nucleotide query
<u>tblastn</u>	Search translated nucleotide database using a protein query
<u>tblastx</u>	Search translated nucleotide database using a translated nucleotide query

INTRODUCTION TO BLAST-II

Smith Waterman can align complete sequences. BLAST does it in an approximate way. Hence, BLAST is faster BUT does not ensure optimal alignment.

Use Cases of BLAST

BLAST provides for approximate sequence matching. **Input** to BLAST is a **FASTA formatted** sequence and a set of search parameters

OUTPUT OF BLAST

Results are shown in **HTML, plain text, and XML formats**

BLAST ALGORITHM

BLAST can search sequence databases and **identify unknown sequences** by **comparing them** to the **known sequences**. This can help identify the parent organism, function and evolutionary history. BLAST performs quick alignments on sequences.

There are two main types of BLAST.

Nucleotides

- **Blastn**: Compares a nucleotide query sequence against a **nucleotide** database.

Proteins

- **Blastp**: Compares an amino acid query sequence against a **protein** database.

There are also many other types of BLAST:

- **Blastx**:

Compares a nucleotide query sequence against a protein sequence database.
Helps find potential **translation** products of unknown nucleotide sequences

•**tblastn:**

Compares a protein query sequence against a **nucleotide sequence** database
Nucleotide sequence dynamically translated into all reading frames

•**tblastx:**

Compares the **six-frame translated** proteins of a nucleotide query sequence against the **six frame translated proteins** of a nucleotide sequence database.

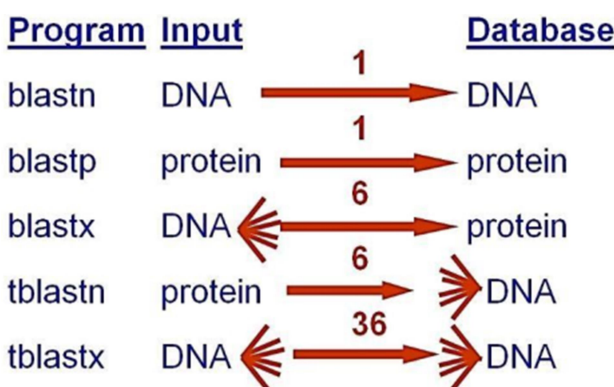
SUMMARY OF BLAST

- BLAST performs **quick** alignments on biological sequences

Step1: obtain a query of sequence

For **known** sequences: Use NCBI, UCSC etc..

For **unknown** sequences: Use NGS or Mass Spectrometry



INTRODUCTION TO FASTA

For comparing **two sequences** we use **pair wise sequencing** and for the comparison of **many sequences** we use **multiple sequence alignment**. To handle the multiple alignments we perform alignment through **smith-waterman algorithm** for **local one**. And for **global alignment** we use **Needleman-wunsch algorithm**.

Both **local and global alignments** are the **dynamic approaches**.

Many of the sequences are compared, which takes time and we use **BLAST** which is an **approximate local alignment search tool**. BLAST compares a large number of sequences, quickly.

FASTA took a similar approach, Developed in **1988** it does **Fast Alignment**. Searches databases for query **protein and nucleotide** sequences. Was **later improved upon in BLAST**.

Online: <http://www.ebi.ac.uk/Tools/sss/fasta/>

FASTA – Fast Alignment Algorithm

Classical global and local alignment algorithms are time consuming. FASTA achieves alignment by using **short lengths of exact matches**.

USES OF FASTA

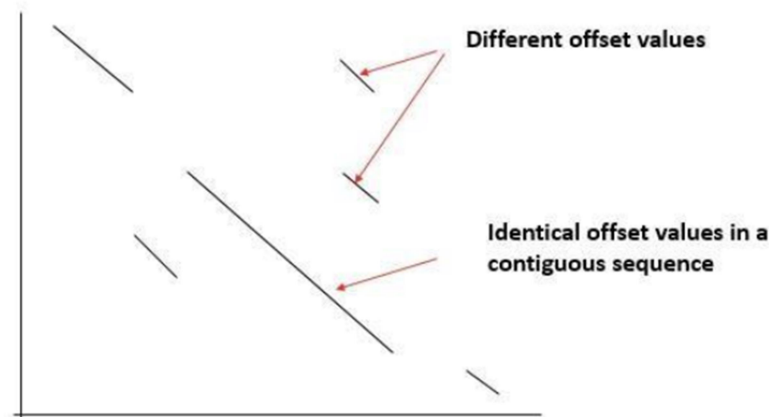
FASTA relies on aligning subsequences of absolute identity. Input to FASTA search can be in FASTA, EMBL, GenBank, PIR, NBRF, PHYLIP or UniProt formats

• OUTPUT OF BLAST

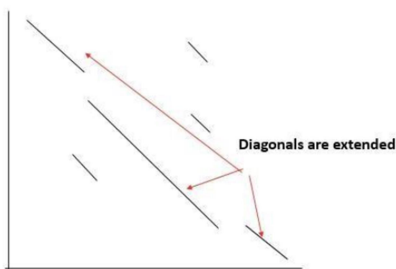
Results are output in **visual format** along with **functional prediction**. Makes table lists the sequence hits found along with scores.

FASTA ALGORITHM

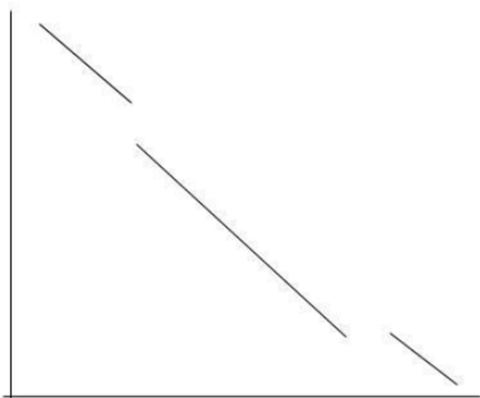
STEP1: Local regions of identity are found



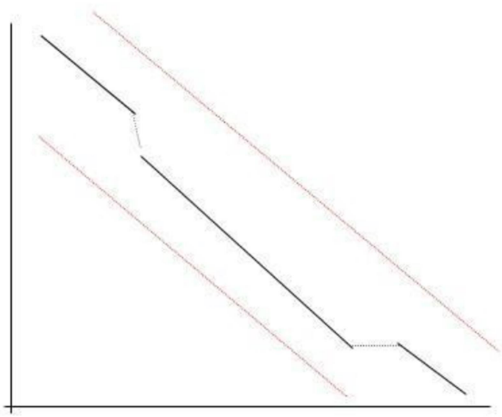
STEP2: Rescore the local regions using PAM or BLOSUM matrix



STEP3: Eliminate short diagonals below a cutoff score



STEP4: Create a gapped alignment in a narrow segment and then perform Smith Watermann alignment



TYPES OF FASTA

There are six types of FASTS:

- **fasts35**
Compare **unordered peptides** to a protein sequence database
- **fastm35**
Compare **ordered peptides** (or short DNA sequences) to a protein (DNA) sequence database
- **Fasta35**
Scan a protein or DNA sequence library for similar sequences
- **Fastx35**
Compare a **translated DNA** sequence (6 ORFs) to a protein sequence database
- **tfastx35**
Compare a **protein sequence** to a DNA sequence database (6 ORFs)
- **fasty35**
Compare a DNA sequence (6ORFs) to a protein sequence

SUMMARY OF FASTA

FASTA can **briskly perform sequence search databases** if given a query sequence.

Multiple types of FASTA exist which assist in aligning DNA/RNA/Protein sequences

fasta35, fasta35_*	scan a protein or DNA sequence library for similar sequences
fastx35, fastx35_t	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
tfastx35, tfastx35_t	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
fasty35, fasty35_t	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
tfasty35, tfasty35_t	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
fasts35, fasts35_t	compare unordered peptides to a protein sequence database
fastm35, fastm35_t	compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database
tfasts35, tfasts35_t	compare unordered peptides to a translated DNA sequence database
fastf35, fastf35_t	compare mixed peptides to a protein sequence database
tfastf35, tfastf35_t	compare mixed peptides to a translated DNA sequence database
ssearch35, ssearch35_t	compare a protein or DNA sequence to a sequence database using the Smith-Waterman algorithm.

http://fasta.bioch.virginia.edu/fasta_docs/fasta35.shtml

BIOLOGICAL DATABASE AND ONLINE TOOLS

Sequences are obtained from **genome sequencing** and **mass spectrometry**

Structures are obtained from **X-Ray Crystallography, Atomic Force Microscopy & Nuclear Magnetic Resonance Spectroscopy**

This sequence and structure data is very useful for researchers

Databases are formed to store and share this data to make biological data available to scientists in **computer-readable form**.

EXPASY

Expaty provides **access to a variety of online databases and tools**

It is developed by **Swiss Bioinformatics Institute (SIB)**. Website provides access to databases and tools. Proteomics, genomics, phylogeny, systems biology, population genetics, transcriptomics etc. can be searched.

<http://www.expasy.org/>

UNIPROT AND SWISSPROT

Both UniProt and SwissProt are the online database for proteins.

Swiss-Prot contains human curated protein information

- Accession number, unique identifier
- The sequence
- Molecular mass
- Observed and predicted modifications

Protein sequences from various species and organisms can be found in uniprot.

SwissProt is the **manually annotated** version of the UniProt Database.

PROTEIN DATA BANK

Protein Data Bank is the premier resource of **protein structures**. These structures have been determined using experimental techniques.

It's Open & Free **Protein Data Bank** provides **Cartesian coordinates** of each **atom** in the **protein structure**. Over **50,000 protein structures** are reported and present in this database.

REVIEW OF SEQUENCE ALIGNMENT

We use **next generation sequencing** and **whole genome sequencing** to obtain the **genetic information**.

For **protein sequencing** we use **Mass Spectrometry** and **Edman Degradation**.

STORAGE:

- Sequence information is stored digitally
- Databases are designed to store sequence data

SHARING AND ACCESS:

- Sequence databases are shared via online websites
- Access to several such websites is free

USAGE OF DATA:

Sequence data can be used to obtain:

- Similarity of sequences

- Evolutionary History
- Predict the function of molecules

GENBANK

- All known **nucleotide sequences** can be obtained from GenBank
- Thousands of **genomes** are available online on GenBank, for free and are searchable
- GenBank: **Public database of nucleotide** sequences for over **200,000 organisms**
- Growing exponentially

Data in GenBank

- Entries can be a **contiguous stretch of DNA or RNA sequences** and their annotations
- Entries are updated **every two months**
- Developed by Swiss Bioinformatics Institute (**SIB**)

<http://www.ncbi.nlm.nih.gov/genbank/>

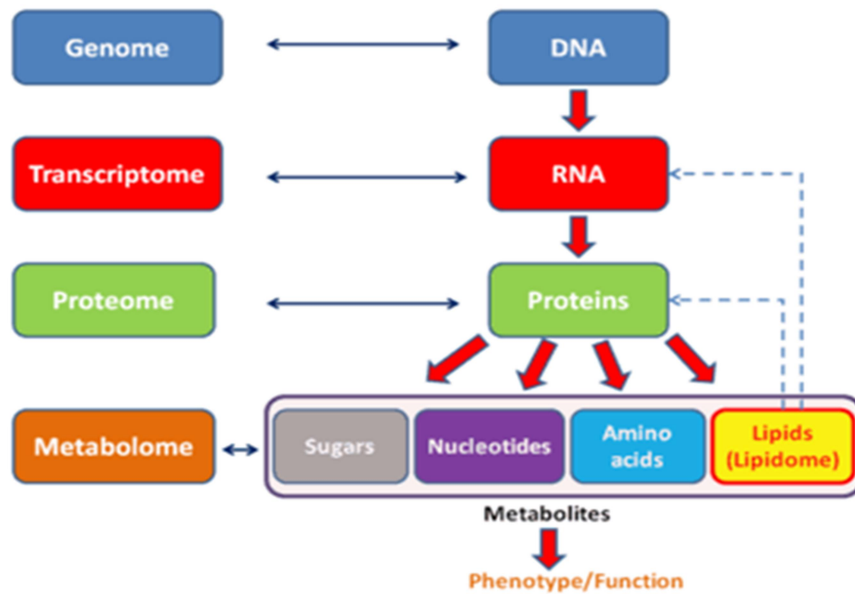
ENSEMBLE

ENSEMBLE is **genome search engine** which is used to search the genome of every recorded species.

<http://asia.ensembl.org/index.html>

Sequence Analysis Overview

Gene, mRNA and Protein Sequences



STORAGE

✓ DNA & RNA

☐ GenBank

- Public database of nucleotide sequences for over 200,000 organisms
- Nucleotide Sequence Database
- Thousands of **genomes**

✓ Proteins

☐ UniProt

Public database of protein sequences

- ✓ **Swiss Prot** (Human Curated Protein Information)

✓ Protein Data Bank

Premier resource of **protein structures**

Over **50,000 protein structures**

✓ Ensemble

Genome search engine

✓ **Pairwise Sequence Alignments**

- Sequence Similarity v/s Identity
- Methods/ types

□ **Dynamic Programming**

To obtain optimal alignments using tracebacks

1. Global Alignment

- Needleman & Wunsch 1970
- To align genes

2. Local Alignment

- Smith & Waterman 1981
- Searching local similarities

✓ **MSA (Multiple Sequence alignments)**

Method for local alignment but no guarantee that answer is best

- FASTA 1988 BLAST 1990

✓ **Progressive MSA** **(CLUSTALW)**

✓ **Scoring Matrices or Substitution Matrices**

- **PAM** **(PAM 120)**
- **BLOSUM** **(BLOSUM 62)**

Chapter 3 - Molecular Evolution

Molecular Evolution & Phylogeny

Molecular evolution is the process of **change in the sequence composition** of cellular molecules such as DNA, RNA, and proteins **across generations**. All molecules have an evolutionary history.

Phylogenetics

Phylogenetics is the study of extracting **evolutionary relationships** between species.

Phylogenetics has led to the **creation of relationship trees** between various species of Bacteria, Archaea, and Eukaryota.

Types of Phylogenetic Trees

Scaled Trees

- Branch lengths are **equal** to the magnitude of change in the nodes

Unscaled Trees

- Only representing the **relationship** between sequences

Evolution of Sequences

DNA acts as **cellular memory unit** and **protein** are the **translated product** of DNA coded information. And evolution is very important to survive in different type of environments.

Method of Change

DNA gets modified by:

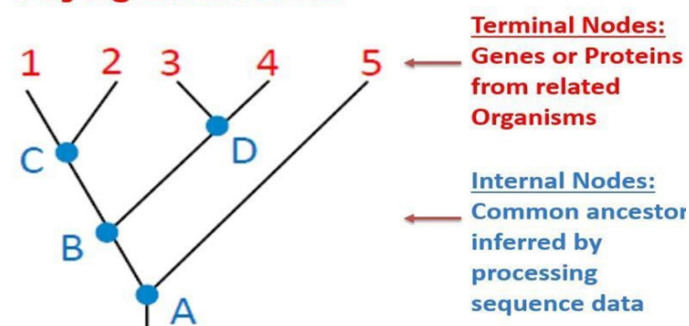
- Mutation & Substitution
- Insertion
- Deletion

The evolutionary events and their combination impart relationships between sequences. These relationships are explored in **Phylogenetics**. Several algorithms exist for finding such relationships

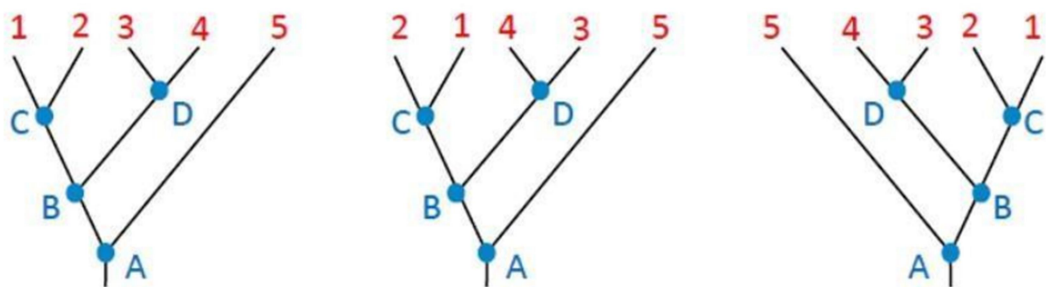
Concepts and Terminologies - I

Phylogenetics involves **processing sequence information** from different species to find **evolutionary relationships**. **Output** from such studies includes **Phylogenetic Trees**.

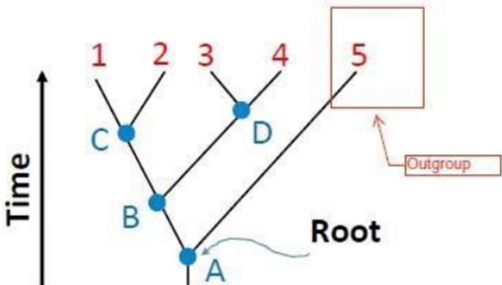
Phylogenetic Trees



In above figure the point A stands for ancestor and with the passage of time the evolution occurred with and the genome sequence of organisms changed.



All trees have same meanings.



Root node is the **ancestor** of all other nodes. The **direction of evolution** is from ancestor to the terminal nodes.

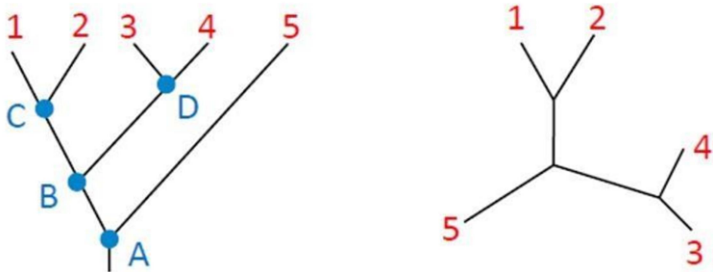
Conclusion

Phylogenetics specifies evolutionary relationship with the help of trees. Trees can be rooted or unrooted. **Rooted trees** can show **temporal evolutionary direction**.

Concepts and Terminologies - II

Rooted and Unrooted trees can be used to show **phylogenetic** relationships between sequences.

Rooted trees are computationally expensive.



Number of Sequences	Number of Rooted Trees	Number of Unrooted Trees
3	3	1
4	15	3
5	105	15
10	34, 459,425	2,027,025
15	213,458,046,767,875	7,905,853,580,625
...	...	...

How many trees?

Rooted: $(2n-3)! / 2^{n-2}(n-2)!$

Unrooted: $(2n-5)! / 2^{n-3}(n-3)!$

Algorithms and Techniques

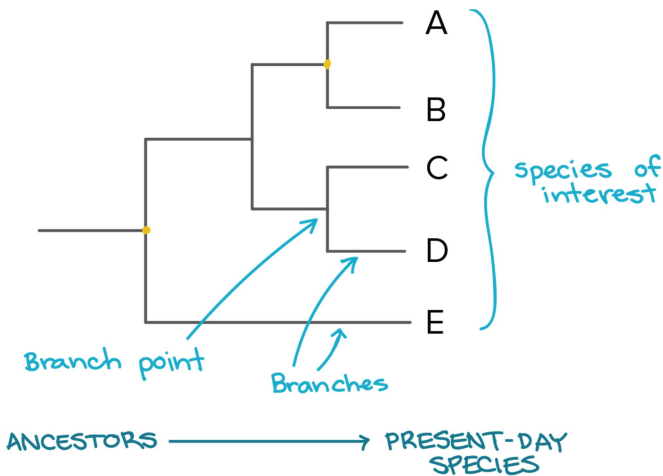
Several types of algorithms exist which are divided into **two classes**. There are many methods for constructing evolutionary trees.

Clustering Approach	Objective based Methods
UPGMA	Least Square Distances
WPGMA	Maximum Likelihood
Neighbor Joining	Maximum Parsimony
Single Linkage	
Complete Linkage	

UPGMA

UPGMA (Unweighted Pair Group Method with Arithmetic Mean) is a simple **agglomerative (bottom-up) hierarchical clustering** method. The method is generally attributed to **Sokal and Michener**.

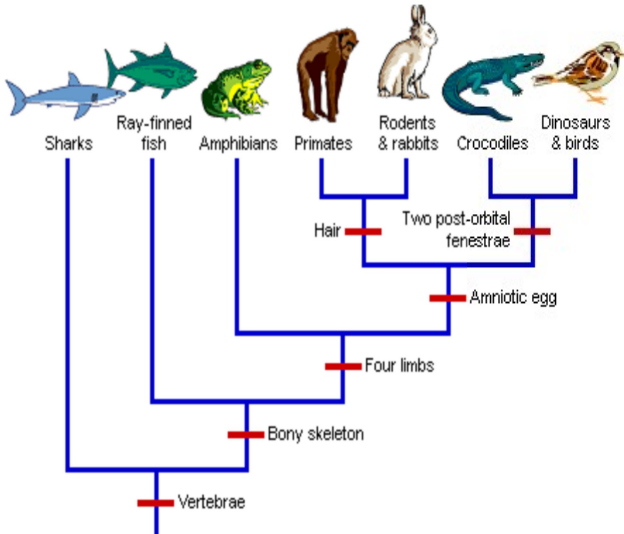
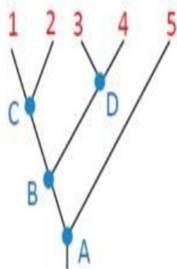
In this method two sequences with the **shortest evolutionary distance** between them are considered and these sequences will be the **last to diverge**, and represented by the **most recent internal node**.



Algorithms and Techniques

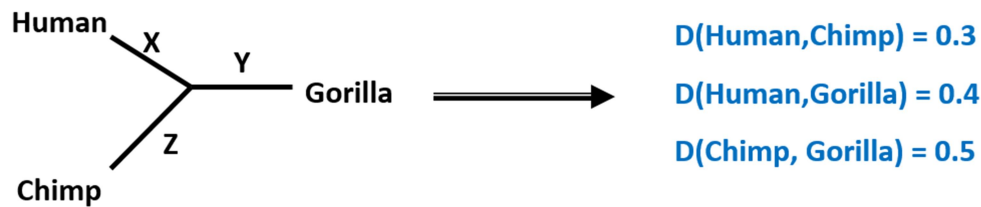
UPGMA: Unweighted Pair - Group Method using arithmetic Averages

- Two sequences with with the **shortest evolutionary distance** between them are considered
- These sequences will be the **last to diverge**, and represented by the **most recent internal node**.



Least Squares Distance Method

Branch lengths, represent the **“observed”** distances between sequences (i & j). Find X, Y and Z such that $D(i, j)$ are conserved?



Conclusion

- Several methods exist for constructing phylogenetic trees.
- Broadly, they belong to **objective methods** or **clustering methods**.
- We will study UPGMA and Distance Methods.

Introduction to UPGMA

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

- Calculating distance between two clusters:

Cluster X + Cluster Y = Cluster Z

Calculate the **distance of a cluster** (e.g. W) **to the new cluster Z**

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

N_x is the **number of sequences** in cluster x

- Calculating distance between two trees:

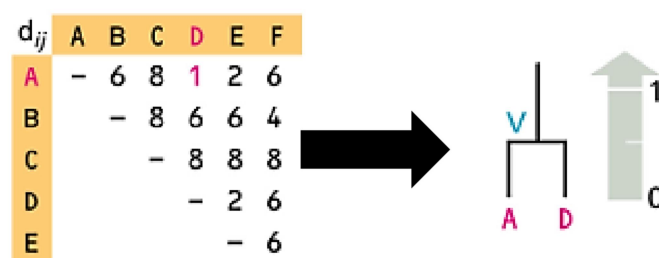
Assume we have N sequences

Cluster X has N_X sequences, cluster Y has N_Y sequences

d_{XY} : the **evolutionary distance** between X and Y

$$d_{XY} = \frac{1}{N_X N_Y} \sum_{i \in X, j \in Y} d_{ij}$$

- **Methods for constructing trees**



The **distance matrix** is obtained using **pairwise sequence alignment**.

A – D becomes a new cluster lets say V. We have to modify the distance matrix. What are the distances between:

- V and B (Calculate),
- V and C,
- V and E,
- V and F.

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1*6 + 1*6}{1+1} = 6$$

UPGMA is a clustering algorithm which can help us compute phylogenetic trees.

UPGMA has **two components** to it. These include **distance calculations** between two clusters and between two trees.

UPGMA-I

UPGMA starts with creating clusters of sequences which are the closest. Next, distance is computed between the new cluster and the remaining sequences. The process is repeated for all sequences

UPGMA - I

Methods for constructing trees

What are the distances between: **V and C (Calculate)!**

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1*8 + 1*8}{1+1} = 8$$

UPGMA - I

Methods for constructing trees

What are the distances between: **V and E (Calculate)!**

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VE} = \frac{N_A d_{AE} + N_D d_{DE}}{N_A + N_D} = \frac{1*2 + 1*2}{1+1} = 2$$

UPGMA - I

Methods for constructing trees

What are the distances between: V and F (Calculate)!

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

$$d_{ZW} = \frac{N_X d_{XW} + N_Y d_{YW}}{N_X + N_Y}$$

$$d_{VF} = \frac{N_A d_{AF} + N_D d_{DF}}{N_A + N_D} = \frac{1 * 6 + 1 * 6}{1 + 1} = 6$$

UPGMA-II

Once a cluster is selected and its distance is computed with all other sequences, we **update** the distance matrix. Next, we select the **shortest distance** from the new matrix and repeat the process.

UPGMA - II

Methods for constructing trees

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

The distance matrix is obtained using pairwise sequence alignment.

UPGMA - II

$$d_{VB} = \frac{N_A d_{AB} + N_D d_{DB}}{N_A + N_D} = \frac{1 * 6 + 1 * 6}{1 + 1} = 6$$

$$d_{VC} = \frac{N_A d_{AC} + N_D d_{DC}}{N_A + N_D} = \frac{1 * 8 + 1 * 8}{1 + 1} = 8$$

$$d_{VE} = \frac{N_A d_{AE} + N_D d_{DE}}{N_A + N_D} = \frac{1 * 2 + 1 * 2}{1 + 1} = 2$$

$$d_{VF} = \frac{N_A d_{AF} + N_D d_{DF}}{N_A + N_D} = \frac{1 * 6 + 1 * 6}{1 + 1} = 6$$

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

UPGMA - II

d_{ij}	A	B	C	D	E	F
A	-	6	8	1	2	6
B		-	8	6	6	4
C			-	8	8	8
D				-	2	6
E					-	6

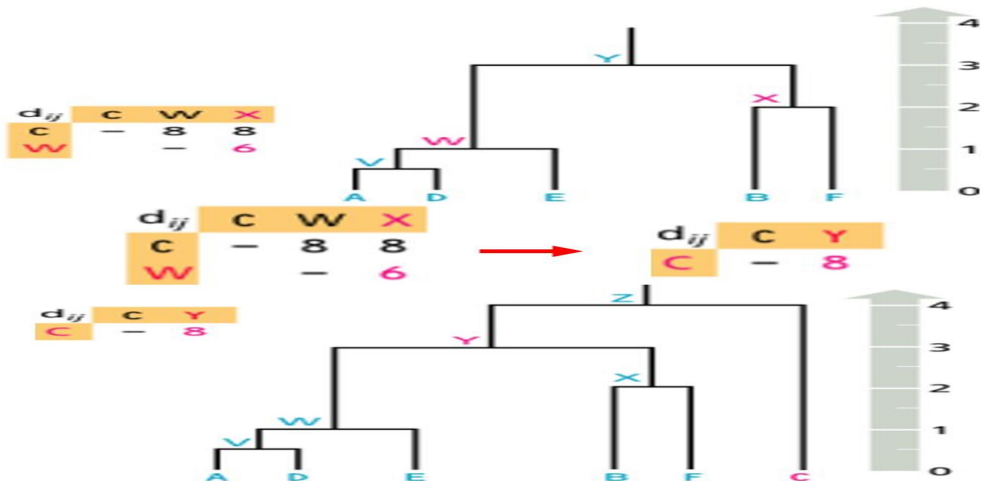
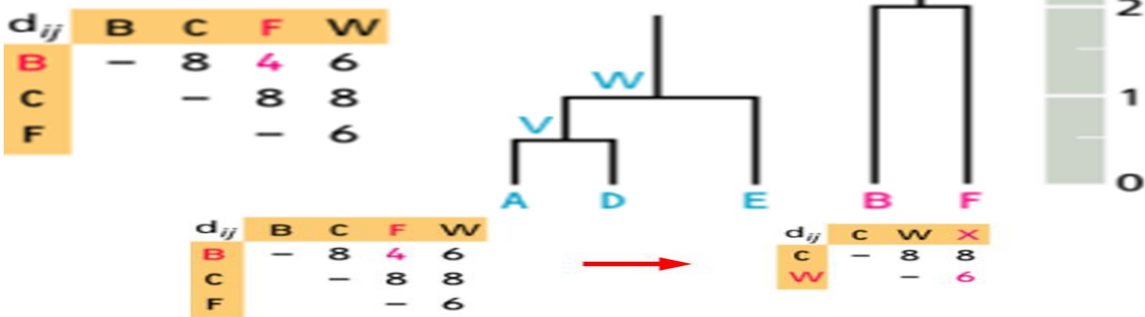
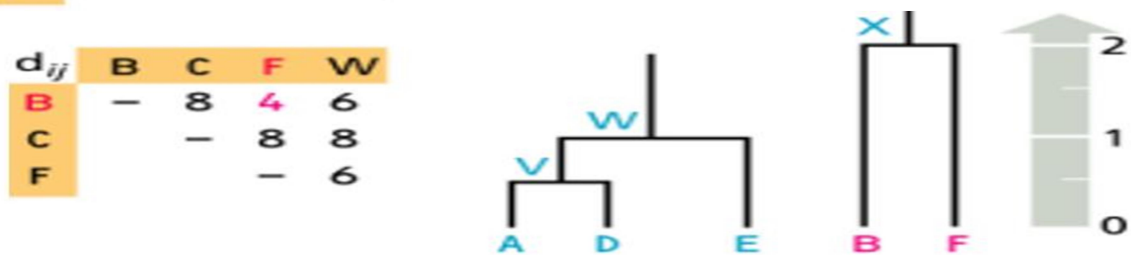
d_{ij}	B	C	E	F	V
B	-	8	6	4	6
C		-	8	8	8
E			-	6	2
F				-	6

- V – E becomes a new cluster lets say W
- Now we have to modify the distance matrix again.
- What are the distances between:
 - W and B,
 - W and C,
 - W and F.

UPGMA - II



- V – E becomes a new cluster lets say W
- Now we have to modify the distance matrix again.
- What are the distances between:
W and B,
W and C,
W and F.



Chapter 4 - RNA Secondary Structure Prediction

DNA TO RNA SEQUENCES

• MOTIVATION

1. In earlier days, RNA was only considered as a **passive intermediary** between DNA and proteins
2. Now we know that **multiple types of RNA** exist e.g. mRNA, tRNA, miRNA and siRNA etc

RNA viruses

Many **viruses** assemble their genomes from RNAs. They are therefore called **RNA viruses**.

Examples include **Human Immunodeficiency Virus** and **Hepatitis C Virus**.

Because **RNA has two (OH) groups** that's why it has **short life span** because of both (OH) **repulsion**

TYPES OF RNA & THEIR FUNCTIONS

There are **two** categories of RNA:

- **Coding RNA**
- **Non-Coding RNA**

Coding RNA performs their **coded function** in protein synthesis. And **Non-coding** RNA helps in **translation process**.

• TYPES OF RNA

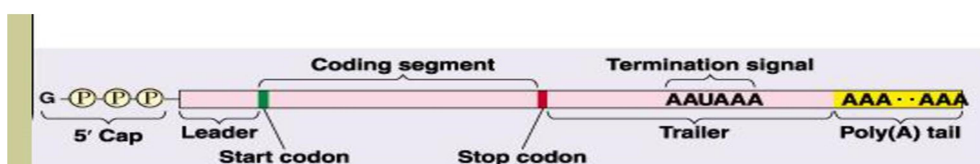
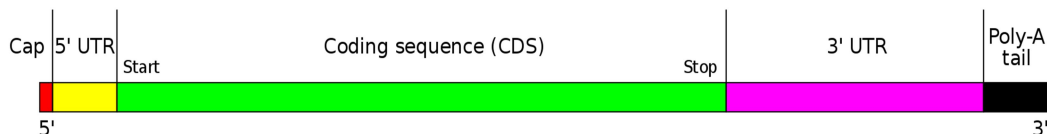
There are many types of RNA according to their functions like:

- **Messenger RNA (mRNA)**
- **Transfer RNA (tRNA)**
- **Ribosomal RNA (rRNA)**
- **Micro RNAs (miRNA)**
- **Small Interfering RNA (siRNA)**

• MESSENGER RNA

1. **Only 5-10%** of this RNA type is **present in cell**.
2. Messenger RNA **5' end** is **capped** with (7-Methyl Guanosine Triphosphate) which helps the **Ribosomes** **to identify** the mRNA.
3. And **3' end** of the mRNA is **poly A tail** (around **30-200 adenylate residues**) which help **shield** against **3' exonucleases**

The structure of a typical human protein coding mRNA including the untranslated regions (UTRs)



■ The **function** of the **5' cap** is:

- prevent mRNA degradation by hydrolytic enzymes
- helps attach to the ribosome

■ **Function** of the **3' tail (Poly A cap)**:

- same functions as the 5' cap
- also helps export of mRNA from nucleus

SIGNIFICANCES OF RNA STRUCTURE

RNA can form **3D structures**, such structural properties helps the **RNA molecule** to perform different functions.

As RNA is composed of **sugars, phosphate and nucleotides** and these nucleotides have ability to form **hydrogen bonds**.

- A' can make hydrogen bonds with 'U'
- 'G' makes hydrogen bonds with 'C'
- 'G' can also make hydrogen bonds with 'U' (**Wobble Pair**)

FUNCTIONS

Due to this ability of bonding RNA forms many structures and due to variety of structures RNA performs many functions in cell like:

- **DNA information transfer**
- **Regulatory roles**
- **Catalytic roles**
- **Defense & immune response**
- **Structure-based special roles**

RNA FOLDING AND ENERGY FOLDING

RNA molecules form many structures for stability and different functions.

A review of free energy

"**Gibbs Free Energy**" (LANGRIDGE and KOLLMAN 1987) is the **free energy** available for RNA molecule for reactions and **RNA structure formation** takes place at this **lower energy**.

Energy of RNA structures

- RNA structures have the **lowest (or close) quantity of free energy**
- In case if RNA has **two structures** we can select the **one with lowest energy state**.

We can **calculate the overall energy** of RNA structures by **summing up energies** given out during the **process of folding**

Calculating energies of structures

- The **stabilizing energy** associated with **stacking base pairs** in a **double-stranded region (-ve)**
- The **destabilizing** influence of unpaired regions (**+ve**)

CALCULATING ENERGIES OF FOLDING-AN EXAMPLE

RNA is composed of **four nucleotides (A, U, C and G)** and these nucleotides are **attached** with **ribose sugar in backbone**. And these nucleotides have **hydrogen bonding** between them. G always bond with C and A always bonds with U through hydrogen bonding and **energy is released**. That's why RNA molecule becomes **more stable**.

5 nucleotides formed **H-Bonds**. This bond formation **released energy (-12.0 kcal/mol)** RNA molecule took up a **2' structure**. Hence became more stable.

TYPES OF RNA SECONDARY STRUCTURES

All the complimentary bases of RNA combine together to **form RNA secondary structures**.

A simple nucleotide sequences of RNA is called as **Primary structure** and **denoted by 1'** while when these nucleotides **fold together and form a complex structure** that is called **secondary structure** and **denoted by 2'**.

The **preferred structure of RNA is 2'** which has many structural patterns like **Helices, Loops, Bulges and Junctions**.

A. Single-stranded RNA

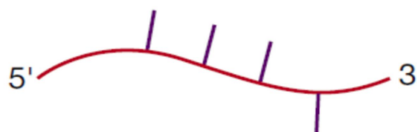


Figure 8 RNA sequence extends from its 5' end to 3' end. Upon folding, 3' end may fold on to the 5' end

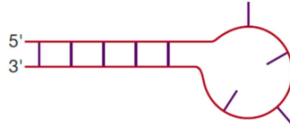
The **first 2' RNA structure** is called **helix**. Unlike the DNA helix, the RNA helix is formed when the RNA **folds onto itself**.

B. Double-stranded RNA helix of stacked base pairs



The **second 2' structure** is the **hairpin loop**.

C. Stem and loop or hairpin loop.



The loop of the hairpin must **at least four bases long** to avoid **steric hindrance** with base-pairing in the stem part of the structure.

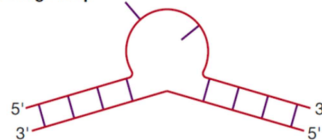
Note: That hairpin **reverses** the **chemical direction** of the RNA molecule

TYPES OF RNA SECONDARY STRUCTURES-II

RNA 1' structure fold the **(5'-3')** ends and make **RNA 2' structure** just like **helix and hairpin structure**.

The **third type of 2' structure** is **bulge loop**.

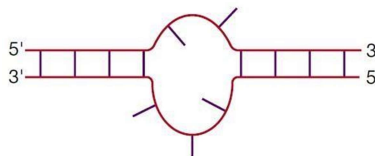
D. Bulge loop



Bulges, are formed when a **double-stranded region cannot form base pairs perfectly**. Bulges can be **asymmetric** with varying number of base pairs on one side of the loop. Bulge loops are **commonly found in helical segments** of cellular RNAs and used to measure the helical twist of RNA in solution. (Tang and Draper 1990)

The **forth type of 2' RNA** structure is **interior loop**.

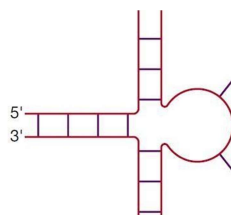
E. Interior loop



Interior loops are formed by an **asymmetric number of unpaired bases** on each side of the loop.

RNA can be fold to form **helices, bulge loops, and interior loops**

The **fifth type of 2' structure** is the **Junction or Intersection**



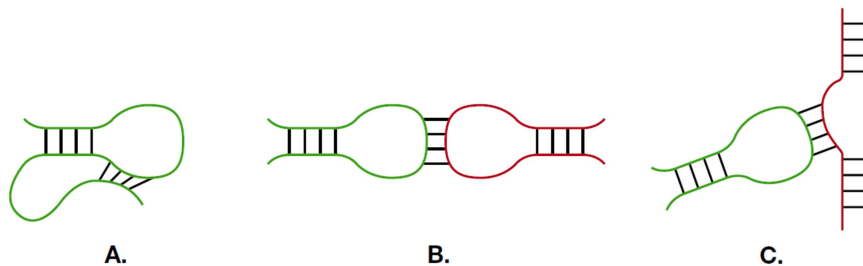
Junctions include **two or more double-stranded regions** converging to form a closed structure.

The **unpaired bases** appear as a bulge

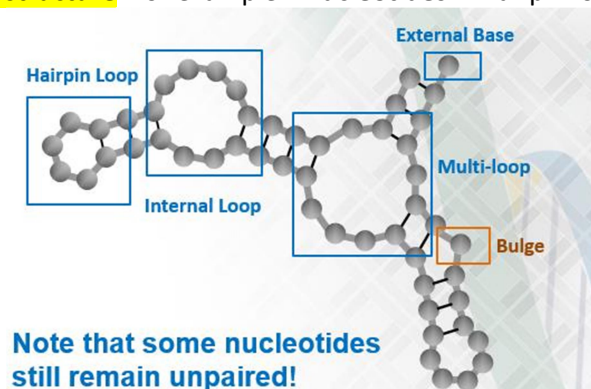
RNA TERTIARY STRUCTURES

RNA tertiary structures are formed when RNA unpaired base bond in 2' region bond.

2' RNA structures is formed due to folding of nucleotide with in RNA molecule but after folding some nucleotides remain open for interaction. And they form hydrogen bonds together.



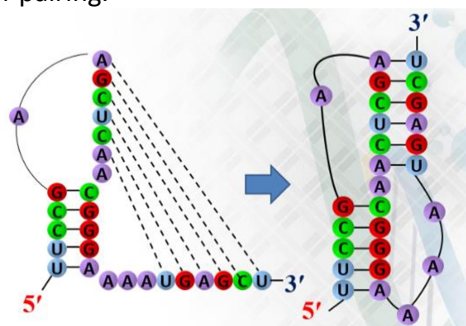
These unpaired nucleotides of 2' structure interact with other unpaired nucleotides and form a third structure called tertiary 3' structure. For example 4 nucleotides in hairpin loop structure does.



The above figure:

1. Indicate how these 2' structures come together
2. Indicate the difference between internal loop and multi loop
3. Indicate the yet unpaired bases

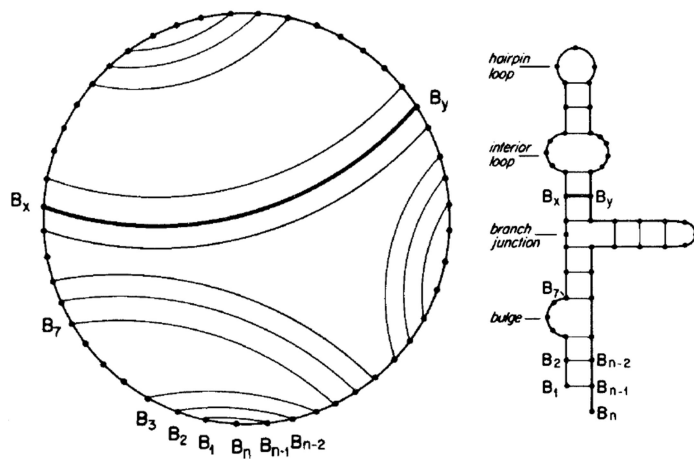
The unpaired bases in 3' structure remain paired by abnormal folding called (pseudoknots) but instead of pairing they remain available or pairing.



- How can we detect pseudoknots in RNA structures?

CIRCULAR REPRESENTATION OF STRUCTURES

Tertiary or 3' structure of RNA may form pseudoknots to detect the pseudoknots in RNA structure we need "circular plot" which is a graphical approach.



Intersecting arcs in circular plot are the pseudoknot.

EXPERIMENTAL METHODS TO DETERMINE RNA STRUCTURES

RNA has 1', 2' and 3' structures. 1' has simple nucleotide sequence and 2' has nucleotides folding and 3' has knots. How can we obtain 2'?

Measuring RNA Structures:

✓ X-Ray Crystallography

Principle: Crystallized RNA diffracts X-rays which helps estimate atomic positions

- Requires RNA crystallization

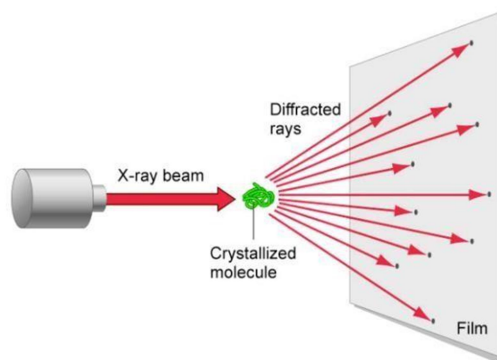
✓ Atomic Force Microscopy

Principle: A laser connected to an Si_3N_4 piezoelectric probe scans an RNA sample

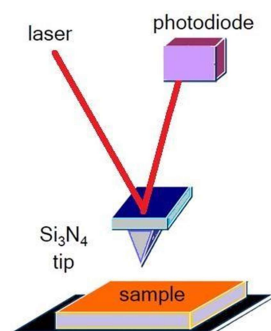
- Works well in air or liquid environment

For measuring the RNA structure we use **X-ray crystallography** (Smyth and Martin 2000), which works according to the principle of diffraction. Crystallized RNA diffracts X-rays which helps estimate atomic positions

All isotopes that contain an odd number of protons and/or of neutrons (see Isotope) have an intrinsic magnetic moment and angular momentum, in other words a nonzero spin, while all nuclides with even numbers of both have a total spin of zero. The most commonly studied nuclei are ^1H and ^{13}C , al



Another method to measure the RNA structure is called as **Atomic Force microscopy** in this technique a laser connected to a Si_3N_4 piezoelectric probe scans an RNA sample. It works well in air and liquid environment.



The third method for measuring the RNA structure is **Nuclear Magnetic Resonance Imaging** in this method Hydrogen atoms in RNA resonate upon placement in a high magnetic field. It Works well without crystallizing RNA

Measuring RNA Structures:

✓ **Nuclear Magnetic Resonance Imaging**

- **Principle:** Hydrogen atoms in RNA resonate upon placement in a high magnetic field
- Works well without crystallizing RNA

• **STORAGE OF STRUCTURES**

Reported structures are stored in online databases. Example includes RNA Bricks and RMDB etc.

RNA Bricks is a database of RNA 3D structure motifs and their contacts, both with themselves and with proteins Stanford University's RNA Mapping Database is an archive that contains results of diverse structural mapping experiments performed on ribonucleic acids.

Strategies for RNA Structure Prediction

RNA structure 2' and 3' can be measured experimentally, but RNA molecule readily degrade due to their short shelf life.

How can we predict 2' and 3' structures:

Give 1' RNA structure creates the 2' structure because the simple nucleotides folds and form 2' structure. And on the base of folding we can predict the stability of the RNA molecule.

Maximizing the number of nucleotides can increase the structure and we have to select the structure according to the stability.

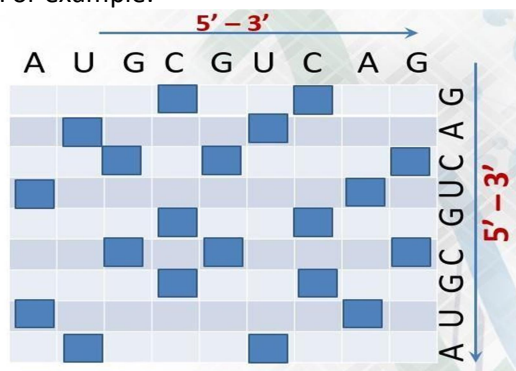
Food for thought:

- Does maximizing nucleotide pairs provides the optimal 2' structure?
- If 2 or more optimal structures are possible, which one should we select?
- Can instability be factored in as well?

Dot Plots for RNA 2' Structure Prediction

Structure measurement through experiments is slow and costly and there is maximum chances of more than one structure existence.

The dot plot method for RNA structure prediction is easy. Draw a square and partition by drawing gridlines. Put RNA sequence on top and left sides of the square. Put a "DOT" on complementary nucleotides For example:



In longest RNA nucleotides the gaps between complementary nucleotides becomes bulges and loops of the structure.

1. ENERGY BASES METHODS

Experimental prediction of RNA structure is slow and costly that's why a few 2' RNA structures are reported experimentally.

While prediction we get many possible 2' structures of RNA and for optimal structure selection we calculate their overall stability.

• **STABILIZING ENERGY**

Energy table helps us to find the optimal prediction of structure because energy is released when complementary nucleotides make bonds.

• **DESTABILIZING ENERGY**

Remaining unpaired nucleotides destabilized the RNA structure in form of hairpin or bulge structure. Example: Eye of the hairpin and bulge

• SUM OF ENERGIES

Sum of stabilizing and destabilizing energies can help determine the quality of a 2' RNA structure.

2' structure with longest coupled sequences vs. one with lowest energy

a. Zuker's Algorithm

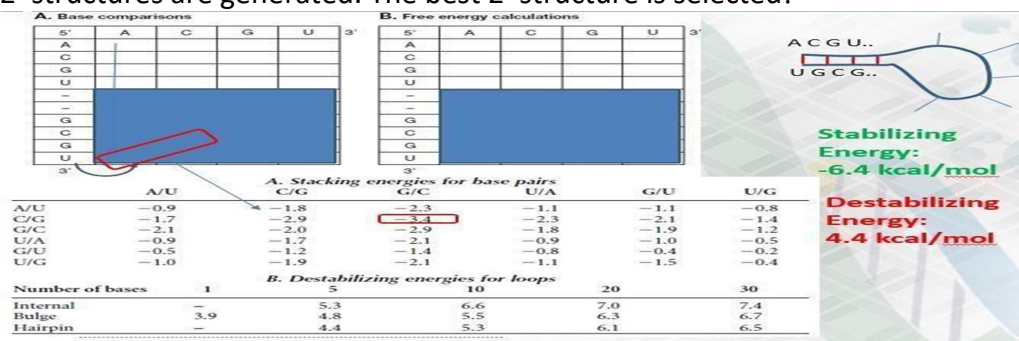
- Compute the stabilizing energies (-ve values)
- Compute the destabilizing energies (+ve values)
- Compute sum of +ve and -ve energies

Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm.

It Compute energies of all possible 2' structures. Generate combinations of all computed 2 structures. Select the one with lowest energy.

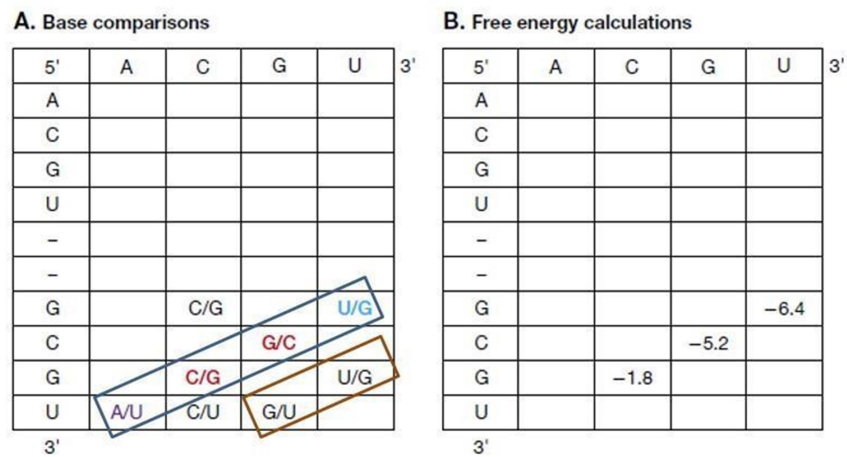
Zuker's Algorithm EXAMPLE

Zuker's Algorithm involves computing stabilizing and destabilizing energies of a 2' structure. All possible 2' structures are generated. The best 2' structure is selected!



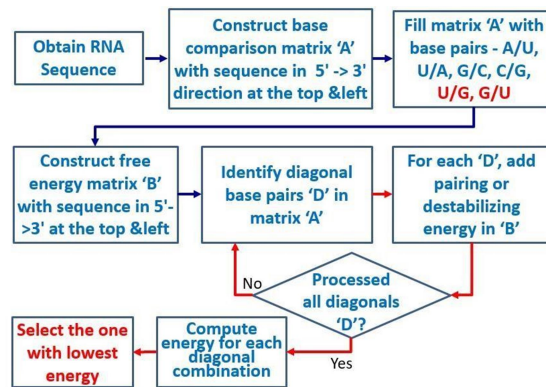
Zuker's Algorithm – A Flow Chart

Zuker's Algorithm involves computing stabilizing and destabilizing energies of 2' structure. And it also computes the overall energy by summing up the positive and negative energies.



The two diagonals ('D') given above include:

1. A/U, C/G, G/C, U/G
2. G/U, U/G



The flow chart for energies

Martinez Algorithm

Important Points:

- Since each 2' structure is weighed by its stability, the optimal structure is quickly shortlisted
- Monte Carlo (MC) simulations do not provide a definitive solution!!

In Martinez algorithm all the 2' structures are weighed by its stability and optimal one is sorted out. Monte Carlo methods (or Monte Carlo experiments) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. They are often used in physical and mathematical problems and are most useful when it is difficult or impossible to use other mathematical methods.

And Monto Carlo method do not provide a definitive solution.

2. Dynamic Programming Approaches

In 2' RNA structure there may be large number of nucleotide sequence with large number of combinations hence it is hard to find the optimal one and for this prediction we use Dynamic Programming (DP) which breaks the larger problems into smaller one.

• PRINCIPLE OF DYNAMIC PROGRAMMING

For optimal structure combination selection we use the Dynamic Programming (DP) and we select the sequence of RNA nucleotides and list all the possible complementary positions for nucleotides in the given complete sequence.

Dynamic Programming then recombines such combinations in a process called "Traceback" to ensure that the highest coupled 2' structure is reported

a. Nussinov An Overview

Nussinov-Jacobson (NJ) Algorithm is a Dynamic Programming (DP) strategy to predict optimal RNA 2' structures, Proposed in 1980. Computes 2' structures with most nucleotide coupling.

HOW IT WORKS

- Create a matrix with RNA sequences on top and right
- Set diagonal & lower tri-diagonal to zero
- Start filling each empty position in matrix by choosing the maximum of 4 scores

Nussinov EXAMPLE

The main points to be focused in N-J Algorithm are:

- Scoring Matrix
- Matrix Initialization

➤ Scoring method

➤ The 4 different positions to be considered for calculating matrix

The matrix is filled by four different positions. **Left, Bottom, Diagonal, and Left/Bottom elements**. In this way all complementary nucleotides coupling is catered.

Score Calculations & Traceback

From four positions the score is calculated and from each position we calculate the score contribution. And maximum score is sorted out.

- There can be multiple tracebacks
- Each traceback can be used to construct an RNA secondary structure
- Select the traceback with the highest number of coupled nucleotides

Comparison of Algorithms

RNA has three different structures 1', 2' and 3'. For these structures predictions there are many algorithms.

But in all algorithm there are two main strategies:

1. Nucleotides stacking

2. Energy minimization

• NUCLEOTIDES STACKING ALGORITHM.

NJ's Algorithm comes under this category. It involves the maximizing the nucleotides pairing. Traceback helps to find best 2' structure.

It **predict** the **75% accurate** 2' structure. Because there may be more than two equal scores as it is calculated from four different positions.

• ENERGY BASED ALGORITHM.

Zuker's Algorithm involves **energy minimization**. It is **updated version** and incorporate the **phylogenetic information**. It is improved. Overcomes the **pseudoknots** assumes them and accommodate them. And this algorithm helps to predict the structures of RNA based on nucleotides.

WEB RESOURCES-I

The mfold web server is one of the oldest web servers in computational molecular biology.

Mfold is **upgraded version of Zuker's algorithm**.

MFOLD is **computationally expensive** and can give results for 1' and 2' structures that have sequences less than **8000 nucleotides**.

Conclusion:

- Given 1' RNA structures, RNA 2' structures can be predicted
- Several algorithms exist for predicting 2' structures
- These include **Zuker, Martinez and NJ** etc.

Online tools

- **MFOLD** – Implements an **upgraded version of Zuker's Algorithm**
- MFOLD outputs a set of possible structures within a given energy range
- It also indicates their reliability

Chapter 5 - Protein Sequences FROM DNA/RNA SEQUENCES TO PROTEIN

A set of **three nucleotides** called **codon**, codes the information **for specific amino acids** in protein synthesis.

Codons select the amino acids and **ribosomes make the protein** by **polymerization process** and these nucleotides coil together to form **3D structure**.

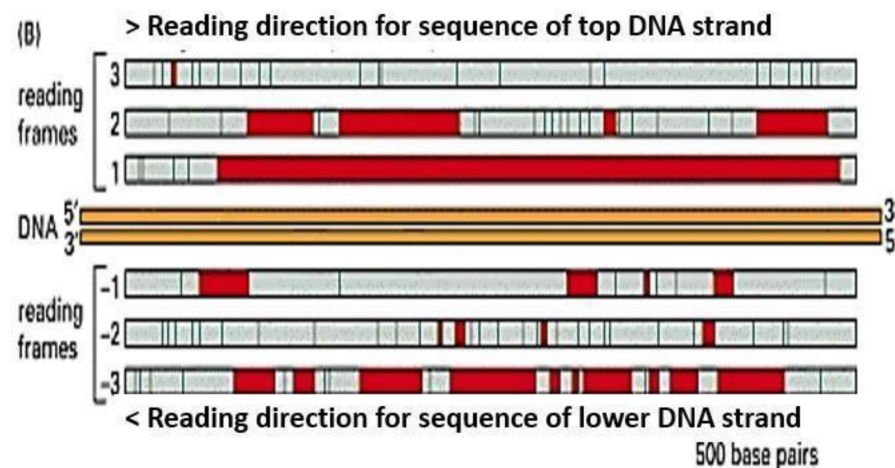
CODING OF AMINO ACIDS

Nucleotides (A, G, C, and T) make set of three called **codons for amino acid selection** in **protein synthesis**. More than one codon can code for same amino acids as there are **20 amino acids** involved in **protein synthesis**.

OPEN READING FRAMES

Codons codes information for amino acid and there are **three stop codons** and **one start codon**. For the **valid open reading** frame it must have **longest sequence**.

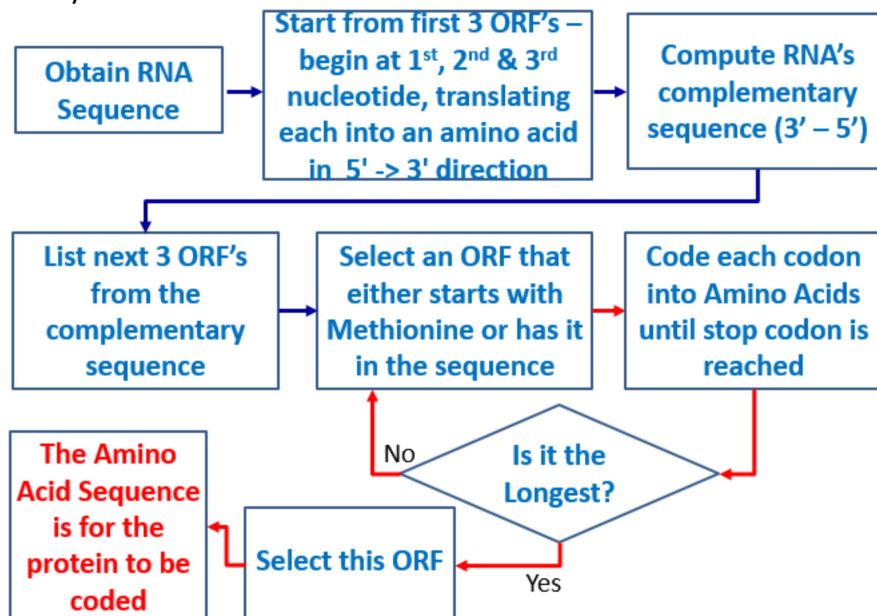
In molecular genetics, an **open reading frame(ORF)** is the part of a **reading frame** that has the potential to code for a protein or peptide. An **ORF** is a continuous stretch of codons that do not contain a stop codon (usually UAA, UAG or UGA).



Six ORF exist in any DNA sequence and longest one is marked and first stop codon will mark the end of the protein.

ORF Extraction – A Flowchart

Selection of ORF is based on its length if it the longest one from others than it would be suitable for protein synthesis reaction.



Both reverse and forward RNA sequences are considered which may have many ORF and selection is based upon longest protein sequences having.

SEQUENCING PROTEINS

Edman degradation, developed by **Pehr Edman**, is a **method of sequencing amino acids** in a **peptide**.

In this method, the **amino-terminal residue** is **labeled** and cleaved from the peptide without disrupting the peptide bonds between other amino acid residues. It starts from the N-terminal and removing **one amino acid at a time**.

Cyclic degradation of peptides by **Phenyl-iso-thio-cyanate (PhNCS)**. PhNCS attaches to the free amino group at N-terminal residue. 1 amino acid is removed as a PhNCS derivative.



• DRAWBACKS

- It is restricted to chain of **60 residues**.
- It is very **time consuming** process 40-50 amino acids per day.

Modern techniques for this is **Tandem mass spectrometry**.

Application of Mass Spectrometry in Protein Sequencing

Protein can be charged with **electrons or protons** and if moving charges are **placed in between** the **magnetic field** **they get deflected**. And their deflection is proportional to their **momentum**.

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \text{ (Lorentz force law)}$$

$$\mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{v}}{dt} \text{ (Newton's second law of motion)}$$

Where:

- **F** is the force applied to the ion
- **m** is the mass of the particle
- **a** is the acceleration
- **Q** is the electric charge,
- **E** is the electric field
- **v × B** is the cross product of the **ion's velocity** and the **magnetic flux density**.

Components of MS

- **Sample Injection**
- **Ionization Source**
- **Mass Analyzer**
- **Ion Detector**
- **Spectra search using computational tools**

CONCLUSION

Charged proteins can be **set into motion** within a **magnetic field**. Their **deflections** accurately correspond to their **molecular mass**. Deflections can be measured (hence protein's mass)

Techniques for MS Proteomics

MS proteomics works on the **principle of protein ionization** which are placed in very **high magnetic field**. Each protein deflects to its proportion which is equal to its molecular weight in this way molecular mass is measured.

The **protein mass of unknown protein** is compared with the masses of proteins in database and matching one is selected.

Example for protein sequences database is **UniProt**, **SwissProt** etc.

Types of MS-based proteomics

Proteins can be sequenced by **Edman's degradation** and **Mass spectrometry**. **MS based proteomics** helps us to sequence the **larger and bigger proteins** **more quickly**.

Following steps are involved in MS:

- Separation
- Ionization
- Mass analysis
- Detection

Two methodologies are involved

1. Bottom up proteomics

2. Top down proteomics

Bottom up proteomics measures the **peptide masses** produced after protein enzymatic digestion.

Top down proteomics measures the **intact proteins** followed by peptides after fragmentation

1. BOTTOM UP PROTEOMICS

In this methodology the protein complex is **treated with site specific enzymes** which cleave (chop) them into amino acid residue and resultant peptides are measured for their masses. **One peptide is selected** at one time for processing and when all are processed than protein search engine is used for matches.

2. TOP DOWN PROTEOMICS

In this methodology **proteins are ionized** and **measured for their masses** and **one protein is mass selected** at a time for fragmentation. And resultant peptide fragments are measured for mass.

We can say that bottom up proteomics deals with peptides while top down proteomics can handle the whole protein.

PROTOCOL

1. Sample containing the mixture of protein from cells and tissues is obtained.
2. Enzymes such as trypsin is use to cleave the proteins.
3. Enzyme cleaves the amino acids at specific sites of amino acid.
4. **Several peptides** are formed when **protein is cleaved**.
5. Number of peptide depends upon the number of sites where enzymes cleaved the protein. For example trypsin cleaves the protein at lysine (k)
6. Mass of each peptide is measured. **Each peptide is measured for its mass (MS1)**
7. One peptide is selected at a time.
8. Different enzyme is use to cleave the protein at different site.
10. **Each resulting peptide is measured for its mass (MS3)**
11. **Each peptide is searched in protein sequence databases for matches**
9. This process keep going until the possible number of peptides are formed or searched.
10. Peptides are searched in data base and matched.

Conclusion

- **Bottom up proteomics** measures the **mass of peptides**
- **Peptides** result after **enzymatic digestion of precursor proteins**
- **Peptides** are searched in **protein databases**

Two Approaches for Bottom up Proteomics (BUP)

There are **two approaches** for **bottom up proteomics**.

1. Peptide Mass Fingerprinting.
2. Shotgun Proteomics

Shotgun Proteomics digest the **whole protein** and mix first and compared with database.

Peptide mass finger printing involves in **protein separation** followed by **single protein's peptide analysis**.

PROTEIN SEQUENCE IDENTIFICATION

Protein Complex
MKLTLMKLSMAIHMSIVHGS
SAMAADSNEK...GAS
GYLPHTPL
ADYLQDQ
TDPHYLDI
KDG
SLKPK
GRFFMGR
EEIEFVQGLNHSTGR
YPEIKAFWPHQEGKDI
LEVLKKYGYTKGDDKV

Filter Protein Database

EST Determination

MS Spectra

Fragmentation MS2

Protein Scores
P62837...51
P62837...51
Q6ZPJ3...40
P21734...33
Q02159...21
P62837...51

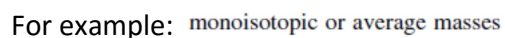
Post Translational Modifications

Matching of Experimental and Insilco Peak List

In Silico Fragmentation of Candidate Proteins

Protein ionization is used in Mass spectrometry based on proteomics protocols. **Ionization involves loading of proton in protein or removal of protein.** Ionizations can **increase or decrease** the mass of protein or peptide.

Is the technique which include **Matrix Assisted Laser Desorption Ionization(MALDI)** & **Electro Spray Ionization (ESI)**



In this technique **one proton is added** to protein or peptide and the molecular weight increases by one and Mass spectrometry reports the **molecule at +1**.

ESI **adds many protons to protein** or peptides and molecular weight is increased by the number of protons added. But it is difficult in ESI to find the molecule with +1.

MS1 & INTACT PROTEIN MASS

Mass/charge helps us to calculate the mass of protein, "**Mass Select**" can help to select **specific MS1** for further analysis.

SCORING INTACT PROTEIN MASS

MS1 helps us to obtain the intact masses of precursor molecules which depend upon the proteomics and protocol applied. Protein masses reported by MS1 are matched with protein database, but before match the masses are converted into +1 of all molecule.

- **SCORING**

We can score each protein in the way that it get maximum score and **low quality matches** should **get low scores**.

After filtering the multiple charges we get the only the peaks having charge 1. And after this filter we compare it with protein data base.

Scoring Intact Protein Mass

Scoring Scheme:

$$M_{Score} = \frac{1}{\sqrt{(M_{Exp} - MT)^2}}$$

Example: A Protein Sequence from Database "MQLF"

MW: 537.67

http://web.expasy.org/compute_pi/

Conclusions:

- Experimental mass reported from **MS1** is matched with theoretical mass of proteins in the database
- **Score** is awarded on the basis of the **closeness between experimental and theoretical mass**

PROTEIN FRAGMENTATION TECHNIQUES

We compare the experimental mass with theoretical data base mass of protein and on base of closeness we rank or score it.

If **several proteins have same score** than selection is done by using another technique **protein fragmentation**. We fragment the protein or peptide and ionize it, it helps us to measure the fragment masses as the same ways as their precursor.

There are different **techniques for protein fragmentation**.

- **Electron Capture Dissociation (ECD)**
- **Electron Transfer Dissociation (ETD)**
- **Collision Induced Dissociated (CID)**

Each fragmentation technique gives result of specific type of fragments.

ECD gives **out 'C' and 'Z' ions**. **CID** gives **out 'B' and 'Y' ions**, etc.

If we can measure the mass of fragments using MS, Calculate the theoretical mass of the fragments. Then, we can award score on the basis of the similarity of experimental and theoretical mass.

TANDEM MS

Tandem MS (or MS/MS, MSn) is a technique **to break down selected ions** (precursor ions) into **fragments** (product ions).

Tandem MS can be extended to the fragments of the intact fragment. All you need is the MS instrument capability to, (i) select fragment's mass range. (ii) Fragment the precursor fragment. Tandem MS helps us to measure masses of fragments. By this scoring and protein identifications so easy.

MEASURING EXPERIMENTAL FRAGMENT'S MASS

In **MS1**, the **molecular weight** of intact sample molecule **is measure** and then **intact molecule** is **fragmented in two afterward**, these **two fragments are measured by MS or MS2**

FRAGMENTATION TECHNIQUES AND MOLECULAR WEIGHT

Fragmentation techniques include ECD, CID etc.

Intact molecule fragmentation splits the molecule into two parts.

FRAGMENT MASS

Mass of fragment is produced by MS2 depending upon the technique because each technique splits the protein or peptide at different location.

Experimental mass reported from MS2 is matched with theoretical peptides of candidate proteins (from DB). Score is awarded on the basis of the closeness between experimental and theoretical masses.